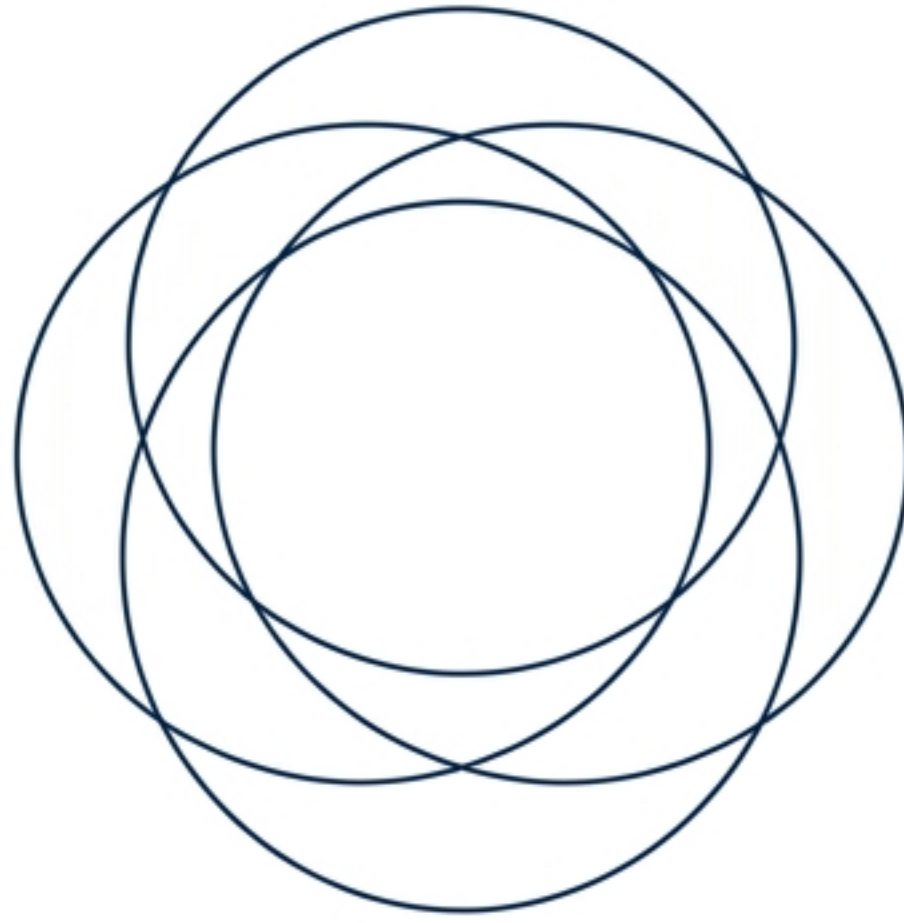




Requirements Engineering

The Architecture of Elicitation and Analysis

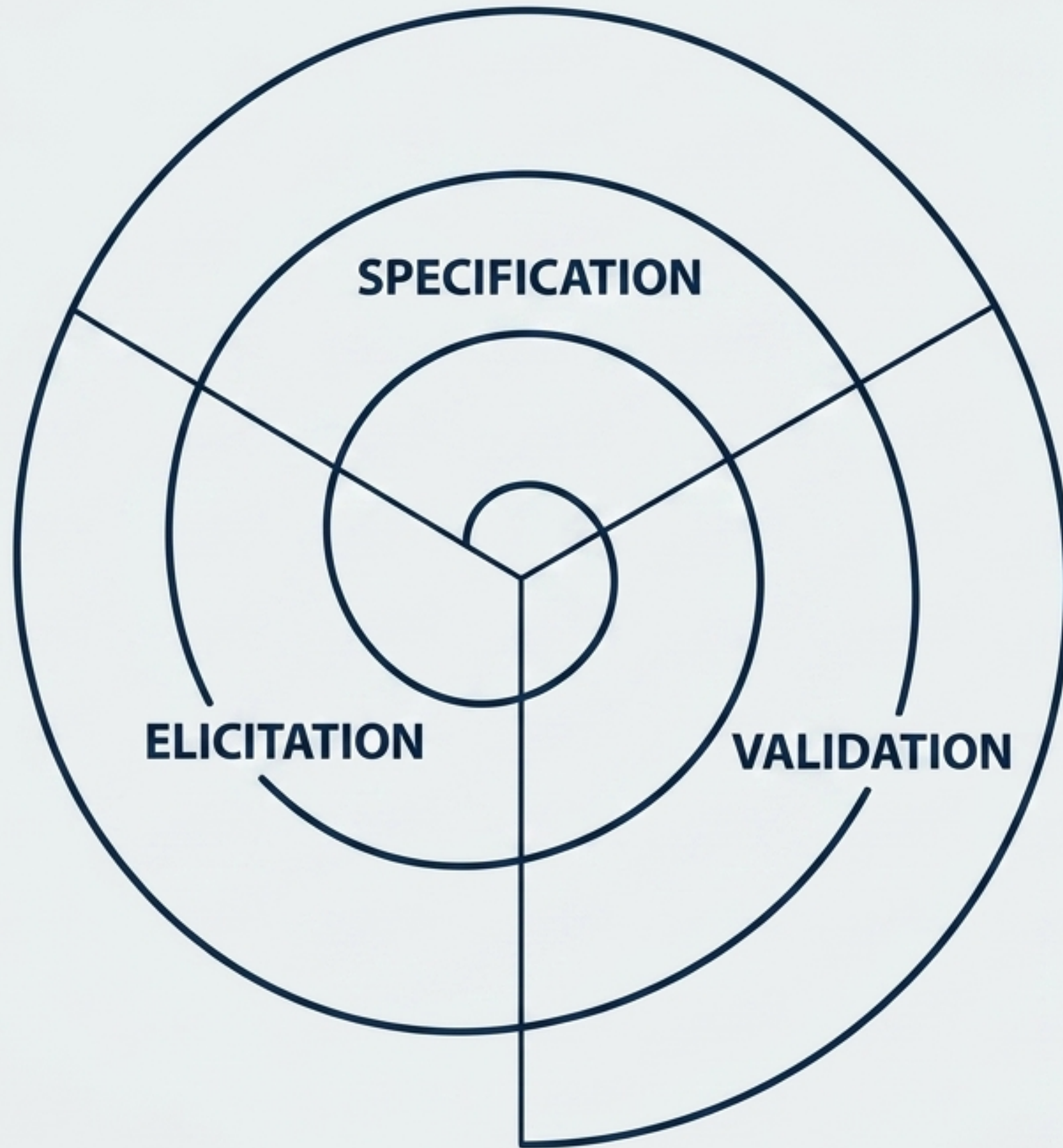
Process, Frameworks, and Techniques



Why Elicitation is Difficult

- Stakeholders cannot clearly articulate their needs.
- Implicit domain knowledge is often left unsaid.
- Diverse stakeholders have conflicting priorities.
- Political power dynamics influence requests.
- The business environment changes mid-analysis.

Requirements are rarely gathered—they must be carefully uncovered.



The Requirements Spiral

Not Sequential: Elicitation, specification, and validation are interleaved.

Inside-Out: Starts with high-level business needs; expands to detailed system constraints.

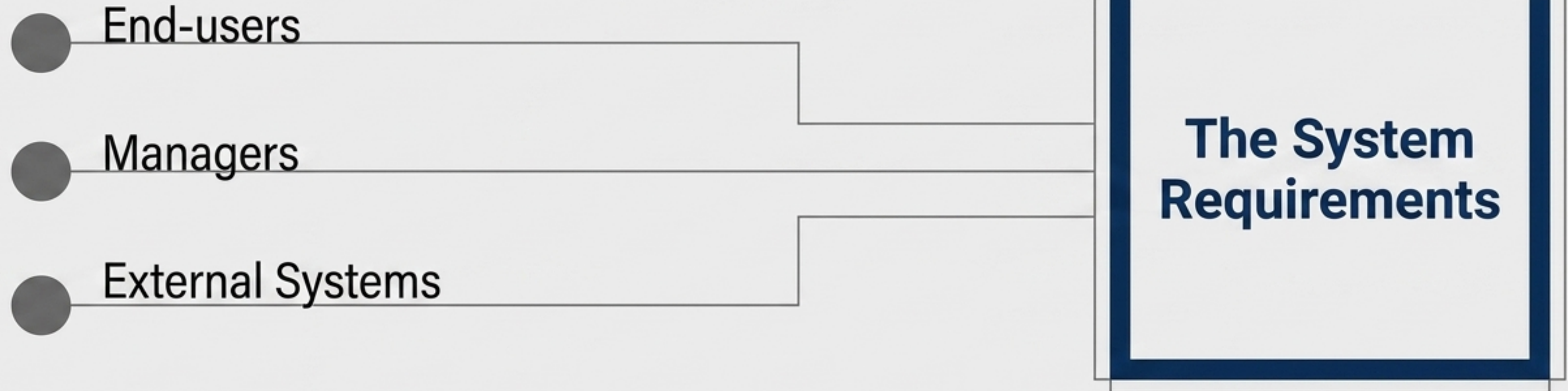
Adaptable: Iterations vary based on methodology (Agile vs. Prototyping).

Requirements engineering is fundamentally iterative.

The Elicitation & Analysis Process



Structuring the Chaos: Viewpoints



Stakeholder information must be actively organized.

Viewpoint Concept: Grouping requirements by stakeholder commonality.

Ensures all voices (users, managers, domain constraints) are represented.

Prevents critical blind spots early in the spiral.

Core Elicitation Techniques



Interviews

Talking to people about their work.



Ethnography

Observing people doing their work.



Stories & Scenarios

Narrating how work could be done.

No single technique is sufficient alone.

The Interview Approach

Closed Interviews: Predefined, structured question sets.

Open Interviews: Exploratory discussions without a strict agenda.

Best Practice: Use a hybrid approach.

Requirement: Always use a springboard (a prototype or specific context) to prompt discussion.

Interviewing Blind Spots

✓ What They Reveal

Overall domain understanding.

Known pain points in current systems.

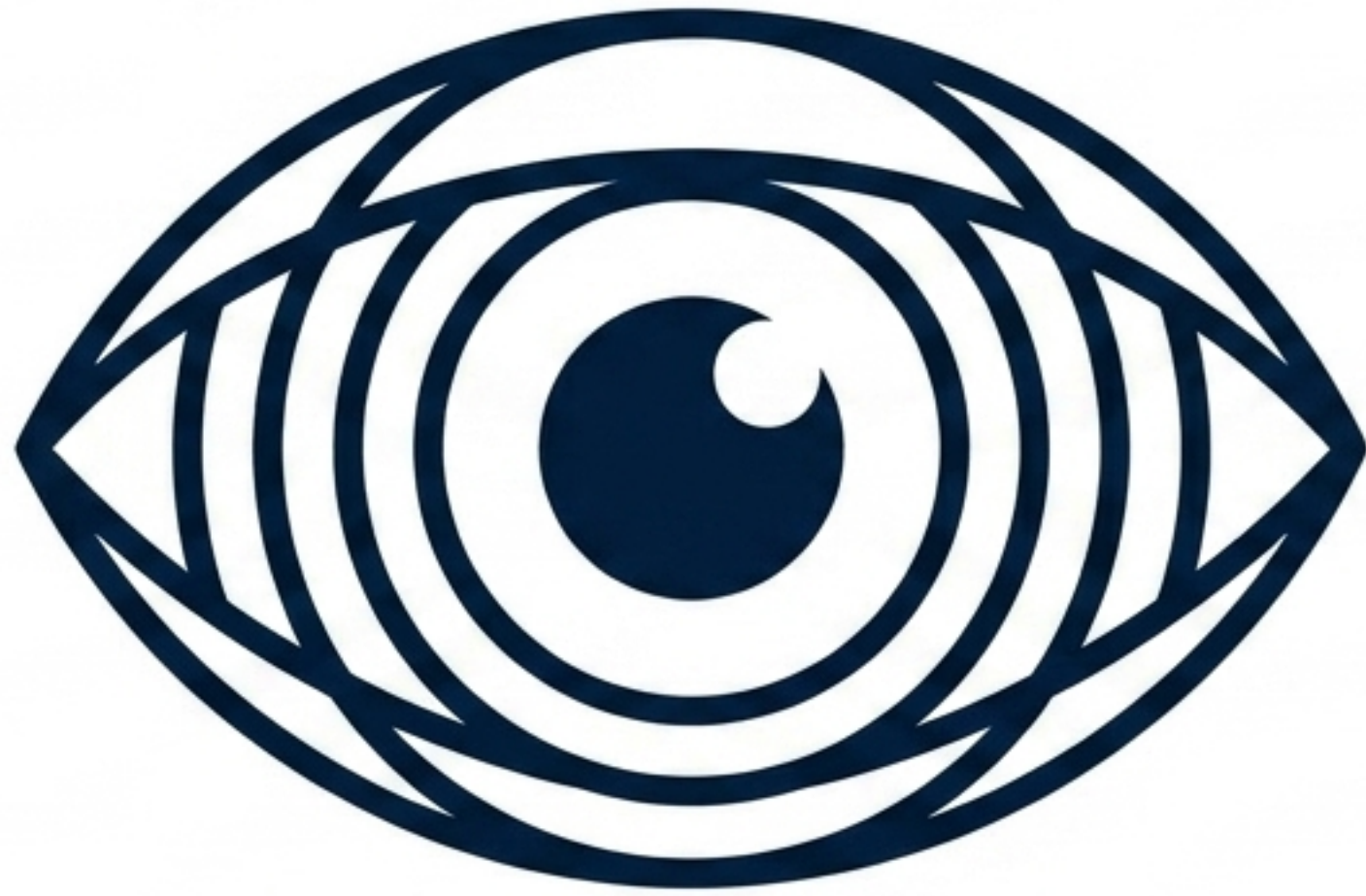
✗ What They Hide

Jargon: Subtle terminology misunderstandings.

The Obvious: Fundamental tasks users forget to mention.

Organizational Politics: Real, unspoken power structures.

Ethnography



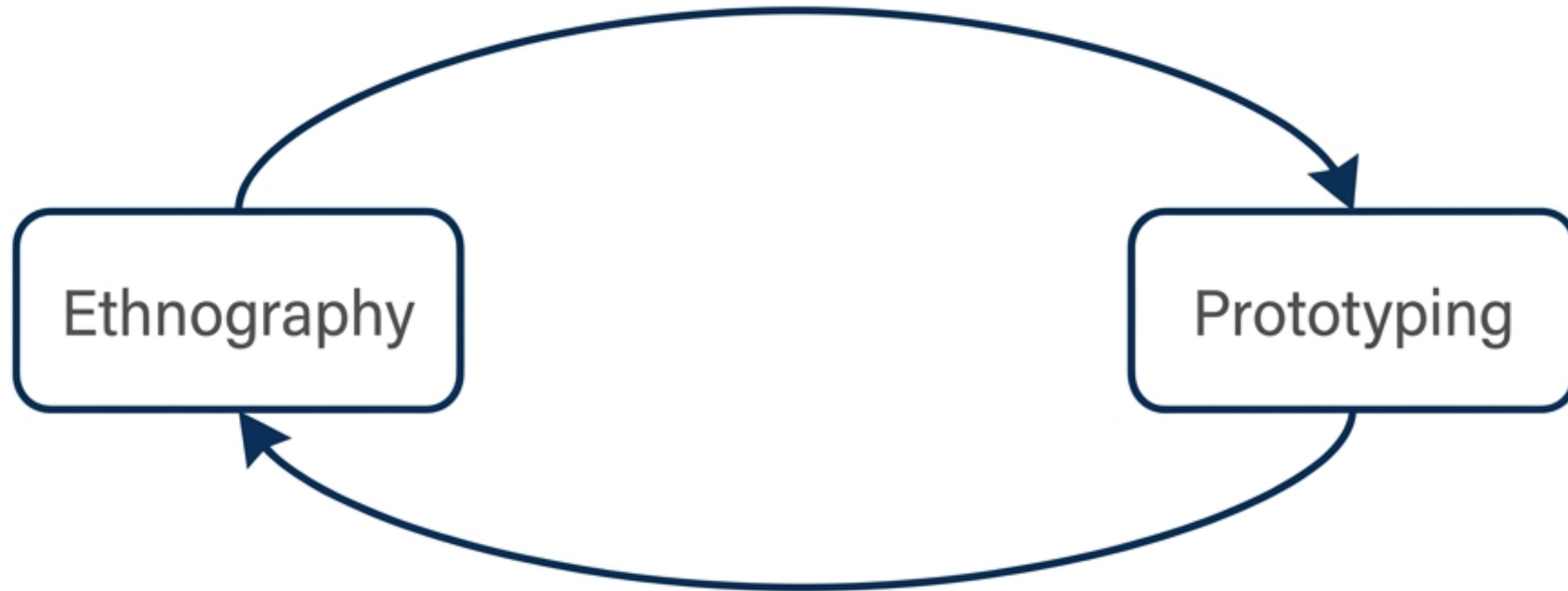
Essential for understanding operational reality.

Immersion: Observing day-to-day work in its actual environment.

Implicit vs. Explicit: Uncovers how people actually work, not how formal processes dictate.

Awareness: Reveals crucial social and cooperative interactions between workers.

Focused Observation



- Ethnography Informs Prototype: Reduces the number of refinement cycles needed.
- Prototype Focuses Ethnography: Identifies specific problems for the observer to investigate next.
- The Limitation: Excellent for existing systems (e.g., Nokia's user studies); poor for radical innovation (e.g., Apple's original iPhone).

Grounding the Abstract



- Stakeholders struggle with abstract system requirements.
- They excel at describing specific, real-life situations.
- Stories: High-level narrative text setting the big picture.
- Scenarios: Highly structured, specific interaction examples.

Comparing Narrative Tools

Stories

Broad, high-level narrative.
Focuses on the Why and Who.
Great for early stakeholder
consensus.

Scenarios

Detailed, structured steps including:

1. Starting state expectations.
2. Normal flow of events.
3. Exception handling (what goes wrong).
4. Concurrent activities.
5. Ending system state.

Elicitation Synthesis

Technique	Primary Strength	Critical Blind Spot
Interviews	General understanding & pain points	Tacit knowledge & unspoken politics
Ethnography	Actual work practices & cooperation	Radical innovation & broader domain rules
Scenarios	Translating human needs to system steps	Can miss high-level business goals

Requirements are not handed to you. They are discovered through iteration, observation, and structured empathy.